

DynCFS: A Program for Modeling Dynamic Coulomb Failure Stress Changes in Layered Elastic Media

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Abstract

Coulomb failure stress change (Δ CFS) quantifies the earthquake-induced difference of shear stress and frictional resistance on a receiver fault, with the latter being proportional to the effective normal stress change. Δ CFS has become a widely used measure for studying earthquake triggering, dynamic rupture processes and earthquake-induced secondary disasters. In simple layered or half-space elastic media, methods for computing static Δ CFS have been well established, with programs such as Coulomb3, PSGRN-PSCMP, and AutoCoulomb being widely used. In contrast, dynamic Δ CFS evaluation generally relies on numerical discretization schemes, such as finite-difference, finite-element, boundary-element and discontinuous Galerkin methods, which, while suitable for complex structures, are computationally expensive. To overcome these limitations, we develop DynCFS, a user-friendly, Green's function based and therefore computationally efficient program for calculating both static and dynamic Δ CFS in layered elastic media. The tool enables rapid assessment of dynamic triggering effects, both between successive earthquakes and among multiple sub-events or faults during an earthquake.

Key words: Computational seismology; Numerical modelling; Earthquake dynamics; Earthquake hazards; Earthquake interaction, forecasting, and prediction

1 Introduction

Since the concept of Coulomb failure stress change (Δ CFS) was introduced by King et al. (1994), it has become a fundamental physical parameter in seismology. Derived from the Coulomb failure criterion, Δ CFS quantifies whether stress perturbations promote or inhibit fault

rupture, and it has been extensively applied to studies of earthquake interaction and triggering mechanisms. Early investigations focused predominantly on static Δ CFS, which represents the permanent stress redistribution following an earthquake and has successfully explained aftershock distributions in numerous cases (King *et al.* 1994; Harris 1998; Stein 1999; Toda *et al.* 2005). Subsequent studies increasingly highlight the importance of dynamic, i.e., fully time-dependent Δ CFS calculations for explaining dynamic triggering, noting that peak dynamic stress amplitudes can far exceed static values in the far-field where triggering can occur with peak dynamic Δ CFS as low as 0.01 MPa (Hill *et al.* 1993; Harris 1998; J. Gomberg *et al.* 1998; Deborah Kilb *et al.* 2000; J. Gomberg *et al.* 2001; Debi Kilb *et al.* 2002; Voisin *et al.* 2004; Freed 2005; Joan Gomberg & Johnson 2005; Hill 2008; Shelly *et al.* 2011; N. Yun *et al.* 2019). Evidence also shows that transient stresses can also trigger diverse modes of fault-slip behaviors, including slow-slip events, tremors, volcanic unrest, and earthquakes at regional to teleseismic distances (Shelly *et al.* 2011; Bayramov *et al.* 2025). Meanwhile, observations of large earthquakes reveal complex, multi-segment rupture processes involving cascading failures, branching and step-overs, and spatiotemporally variable source mechanisms (Vallée *et al.* 2008; Y. Zhang *et al.* 2009; Schwartz *et al.* 2012; H. Zhang *et al.* 2017; Ruppert *et al.* 2018; Ulrich *et al.* 2019; Wollherr *et al.* 2019; Y. Xu *et al.* 2022; C. Ren *et al.* 2024; Stein & Bird 2024; J. Zhou *et al.* 2025). Together, these insights motivate an explicitly time-dependent view of fault interactions, for which dynamic Δ CFS offers a natural framework to quantify wave- and rupture-front-mediated loading among faults.

Practical tools for computing static Δ CFS in homogeneous half-space or layered elastic media have been well-established and widely adopted in routine seismological analyses. Programs such as Coulomb3 (Toda *et al.* 2011), PSCMP-PSGRN (R. Wang *et al.* 2006), and AutoCoulomb (J. Wang *et al.* 2021) are current standard resources for static stress modeling. Among these, Coulomb3 and AutoCoulomb utilize an elastic and uniform half-space model, whereas PSCMP-PSGRN is capable of calculating stress in a viscoelastic and elastic layered half-space. In contrast, dynamic Δ CFS calculations require the computation of time-dependent stress fields from prescribed spatial-temporal rupture processes. The requisite seismic wavefields are typically obtained through sophisticated numerical approaches, including finite-difference, finite-element, boundary-element, or discontinuous-Galerkin methods. While these

61 methods demonstrate excellent performance in fully three-dimensional heterogeneous media,
62 they are computationally expensive, which limits their practical application to routine stress
63 calculations. Alternative methods based on Green's function approaches have been developed
64 specifically for layered half-space models. Early implementations utilizing the original
65 Thomson–Haskell propagator algorithm (Thomson 1950; Haskell 1953; Kennett & Kerry 1979;
66 Cotton & Coutant 1997) encountered several numerical difficulties, including precision
67 degradation for evanescent waves (R. Wang 1999) and contamination of far-field oscillations
68 by smooth near-field spectra (R. Wang & H. Wang 2007). These limitations were systematically
69 addressed and resolved through a series of methodological advances by R. Wang et al. (1999,
70 2007, 2017), employing the orthonormalized propagation matrix method and differential
71 transformations.

72 Building upon these theoretical and computational developments, we employ the EDGRN–
73 EDCMP program (R. Wang 2003) to construct Green's function libraries for static stress
74 changes, and a modified version of QSEIS (R. Wang 1999; R. Wang & H. Wang 2007) or QSSP
75 (R. Wang *et al.* 2017) to compute dynamic-stress Green's functions. Integrating these robust
76 computational components, we develop DynCFS, an open-source Python-based software tool
77 specifically designed for computing complete Δ CFS in layered Earth models. Our program can
78 compute Δ CFS on (i) receiver faults with a prescribed mechanism, (ii) receiver faults with fixed
79 geometry but rake optimized with respect to a regional tectonic stress field, and (iii) receiver
80 faults for which both geometry and rake are optimized under a tectonic stress field. The program
81 features streamlined installation procedures, optimized computational efficiency through
82 vectorized operations and parallel processing capabilities, providing a convenient and rapid
83 evaluation of both static and dynamic Δ CFS. We apply the program to the 2008 M_w 7.9
84 Wenchuan and 2014 M_w 6.1 Ludian earthquakes and obtain the dynamic Δ CFS results on a
85 specified fault plane as well as on a plane at a fixed depth.

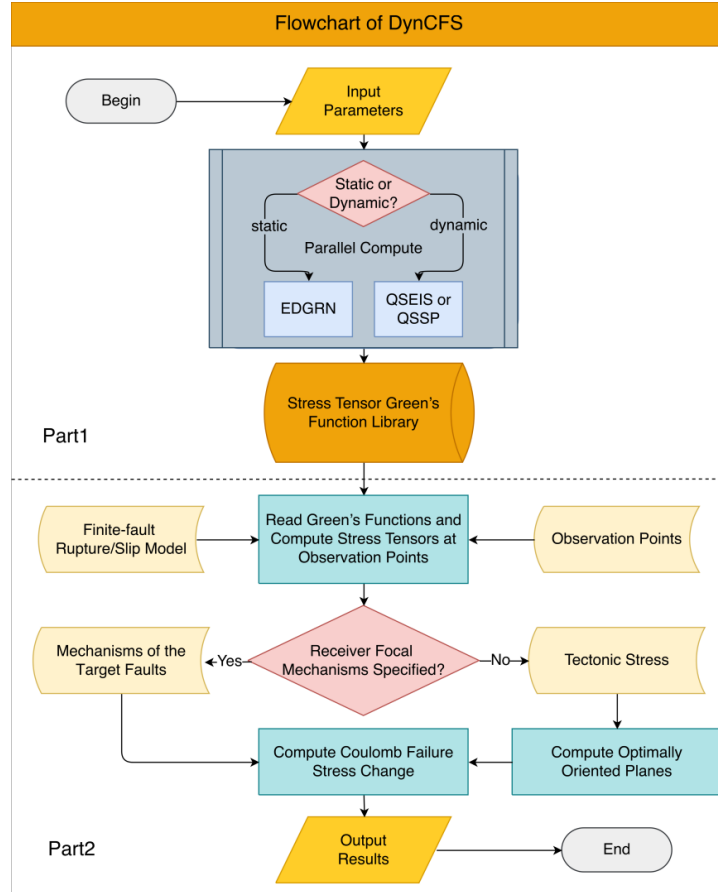


Figure 1. Flowchart of the DynCFS program.

2 Method and Workflow

The flowchart of our program is illustrated in Fig. 1 and can be broadly divided into two parts. One part involves constructing a Green's function library of stress tensors, and the other retrieves stress tensors from the library to compute Δ CFS.

2.1 Build Stress Tensor Green's Function Library

Our computational approach employs different strategies for static and dynamic stress calculations, optimized according to the spatial scale and accuracy requirements. For static stress, we construct the Green's function library using EDGRN (R. Wang 2003), a program designed to compute static stress in layered elastic half-space media. Because static Δ CFS values are generally small in the far field, it is unnecessary to consider spherically layered Earth models. Dynamic stress calculations are performed using either QSEIS (R. Wang 1999; R. Wang & H. Wang 2007) or QSSP (R. Wang *et al.* 2017), depending on the epicentral distance. For near-field calculations, we employ the QSEIS program, which utilizes a layered half-space media model. Since the original QSEIS program outputs only displacement or velocity time

series, we modified the codes to calculate stress and stress-rate tensors through analytical differentiation in the wavenumber domain (see Appendix 1 for details). The choice of QSEIS for near-field applications is motivated by several key advantages. In the near-field region, the Earth's spherical geometry can be well approximated by a flat layered model, enabling QSEIS to provide rapid and precise results. Furthermore, near-field stress distributions exhibit steep spatial gradients that require high spatial resolution. To achieve comparable accuracy with programs that employ spherical layered models (e.g., QSSP), extremely high harmonic degrees are necessary to satisfy the required spatial resolution, resulting in much more computational time than QSEIS. For far-field calculations where the effect of Earth's spherical geometry becomes significant, we utilized QSSP to build the stress Green's function library, which is specifically designed for computing synthetic seismograms in spherically layered media. Because Green's function calculations at different source depths are mutually independent, we parallelized the library construction and implemented two parallelization schemes to accelerate the workflow: (1) multi-process parallelization on a single node (personal computer) and (2) MPI-based parallelization across multiple nodes on a high-performance computing cluster. For a set of specified epicentral distances, we computed the ten non-zero stress components associated with several fundamental moment tensors (Appendix 1, Equations A12.1–A12.4) to build the Green's function library, while the special case of zero epicentral distance was handled separately (Appendix 1, Equations A18.1–A18.9). If a Green's function library for the same Earth model at the required numerical accuracy already exists, we proceed directly to the next stage of the calculation.

2.2 Compute Coulomb Failure Stress Change

We define ΔCFS after King *et al.* (1994), adopting a tension-positive convention and taking the Biot coefficient to be unity ($\alpha = 1$). Thus

$$\Delta\text{CFS} = \Delta\tau + \mu \cdot (\Delta\sigma_n - \Delta p), \quad (1)$$

where $\Delta\tau$ is the shear stress change resolved in the receiver-fault slip direction, $\Delta\sigma_n$ is the normal stress change (positive for unclamping, i.e., a decrease in compressive normal stress), Δp is the pore-fluid pressure change (p denotes the pore-fluid pressure; while $p \geq 0$ in absolute terms, its perturbation Δp may be positive or negative), and μ is the coefficient of friction. In a homogeneous isotropic poroelastic medium, the pore pressure change under undrained

conditions is $\Delta p = B \cdot \Delta \sigma_{kk} / 3$ (Rice & Cleary 1976; Roeloffs 1996), where B is the Skempton's coefficient and $\Delta \sigma_{kk}$ is the sum of the diagonal elements of the stress change tensor (i.e., the volumetric stress change). If we adopt the assumption that the pore pressure change is proportional to the normal stress change $\Delta p \sim \Delta \sigma_n$, we can employ the apparent friction coefficient (Harris 1998; Beeler *et al.* 2000) $\mu' = \mu \cdot (1 - \Delta p / \Delta \sigma_n)$. If we use the Rice model, where $\Delta \sigma_{kk} / 3$ equals $\Delta \sigma_n$, then we have $\mu' = \mu \cdot (1 - B)$. In this case, we can rewrite Equation (1) as (King *et al.* 1994; Harris 1998; Stein 1999; Beeler *et al.* 2000)

$$\Delta CFS = \Delta \tau + \mu' \cdot \Delta \sigma_n. \quad (2)$$

The established Green's function library for full stress tensor can be directly applied in calculations of Equations (1) or (2). From the Green's function library, the stress components corresponding to a specified combination of source depth, receiver depth, and epicentral distance can be retrieved and combined by rotational superposition to synthesize the stress change tensor $\Delta \sigma_e$ for a given moment tensor solution at any azimuth (Appendix 1, Equations A13–A14). When the field point does not coincide exactly with a grid point in the Green's function library, we apply trilinear interpolation using the two nearest source depths, the two nearest receiver depths, and the two nearest epicentral distances. Once the stress tensor is obtained, the traction on the receiver fault plane can be computed as

$$\Delta \mathbf{T}_n = \Delta \sigma_e \cdot \mathbf{n}, \quad (3)$$

where $\mathbf{n}$ is the unit normal vector of the receiver fault plane, and $\Delta \mathbf{T}_n$ is the traction change on the receiver fault plane. A subsequent vector dot product along $\mathbf{n}$ and the unit slip vector $\mathbf{d}$ yields the normal and shear stress components defined in Equations (1) and (2).

$$\Delta \sigma_n = \Delta \mathbf{T}_n \cdot \mathbf{n}, \quad (4.1)$$

$$\Delta \tau = \Delta \mathbf{T}_n \cdot \mathbf{d}. \quad (4.2)$$

The procedure described above corresponds to calculating ΔCFS on a specified plane with fixed focal mechanism. Given a tectonic stress field, evaluating the ΔCFS on the optimally oriented plane (OOP) is necessary. Although actual earthquakes preferentially occur on pre-existing faults, the OOP analysis remains useful as a theoretical limit or for estimating potential hazard in regions where specific fault structures have not yet been identified. The OOP is defined as the plane most susceptible to rupture according to the Mohr-Coulomb failure criterion (King *et al.* 1994), that is, the fault plane that maximizes Coulomb failure stress (CFS) after an

earthquake. The definition of CFS is

$$\text{CFS} = \tau + \mu \cdot (\sigma_n - p), \quad (5)$$

where τ is the absolute shear stress change resolved in the receiver-fault slip direction, and σ_n is the absolute normal stress. For a given stress change $\Delta\sigma_e$ induced by an earthquake, and a background tectonic stress σ_t at the field point, the full stress tensor is

$$\sigma = \Delta\sigma_e + \sigma_t. \quad (6)$$

The three eigenvalues σ_1 , σ_2 , and σ_3 (ordered from minimum to maximum) of σ represent the principal stresses, and the corresponding eigenvectors $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ indicate the orientations of the principal stress axes. The values of σ_1 , σ_2 , and σ_3 can be used to construct a three-dimensional Mohr's Circle representation in the τ - σ_n domain, as shown in Fig. 2.

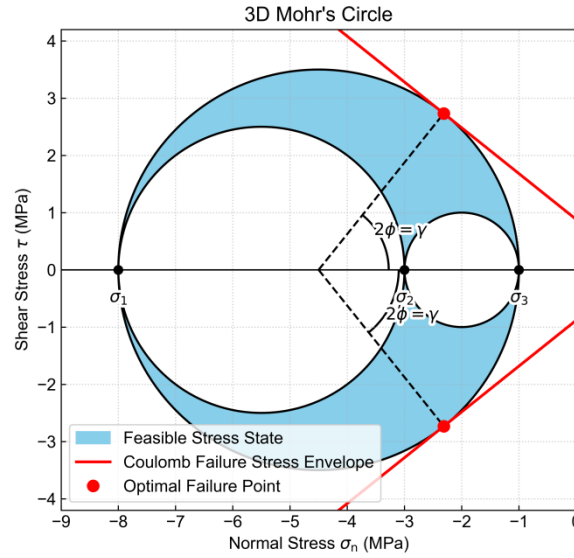


Figure 2. Three-dimensional Mohr's Circle with the feasible stress state (skyblue areas), Coulomb failure stress envelope (red lines) and the points most susceptible to fail (red dots).

Regarding the upper half of the Mohr's Circle, since any feasible stress state point inside the circle defined by σ_1 , σ_2 and σ_3 can always be shifted vertically upward to achieve a larger shear stress change τ (and thus a larger CFS), the optimal failure point must lie on this circle (represented by the red dot in Fig. 2). And for the lower half of the Mohr's Circle, the result is symmetric. Rewriting the shear and normal stress components on this Mohr's Circle gives

$$\tau = r \sin(2\phi), \quad (7.1)$$

$$\sigma_n = r \cos(2\phi) + \sigma_0, \quad (7.2)$$

where $r = (\sigma_3 - \sigma_1)/2$, $\sigma_0 = (\sigma_3 + \sigma_1)/2$ and ϕ is the angle between the normal vector of the resolved plane and $\mathbf{n}_3$. Combining Equations (5) with Equations (7.1)-(7.2), the Coulomb failure stress can then be written as

$$\text{CFS} = r \sqrt{\mu^2 + 1} \cos(2\phi - \gamma) + \mu (\sigma_0 - p), \quad (8)$$

with $\gamma = \arctan(1/\mu)$. Equation (8) shows that the maximum CFS is attained when $2\phi = \gamma$. By symmetry, when τ is reversed in sign, another plane with orientation $2\phi = -\gamma$ yields the same maximum CFS, but with a different rupture direction. Hence, the two angles between the normal vector of OOP and $\mathbf{n}_3$ are

$$\phi = \pm \frac{1}{2} \arctan\left(\frac{1}{\mu}\right). \quad (9)$$

Since p does not affect the value of ϕ , it can be neglected when determining the OOP. It should be emphasized that μ , rather than μ' , should be used here, even when Equation (2) is adopted to compute ΔCFS . This is because the statement preceding Equation (2) assumes only that Δp is proportional to $\Delta\sigma_n$, rather than p being proportional to σ_n , although the latter assumption has been used in some procedures such as Coulomb3. By assembling the eigenvectors $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ as columns to form a rotation matrix and multiplying it by the normal vector of the OOP in the principal stress coordinate system, the unit normal vector $\mathbf{n}_{\text{OOP}}$ in the geographic coordinate system can be obtained

$$\mathbf{n}_{\text{OOP}} = [\mathbf{n}_3, \mathbf{n}_2, \mathbf{n}_1] \cdot [\cos\phi, 0, \sin\phi]^T. \quad (10)$$

And the unit slip vector $\mathbf{d}_{\text{OOP}}$ can be calculated by

$$\mathbf{D}_{\text{OOP}} = \boldsymbol{\sigma} \cdot \mathbf{n}_{\text{OOP}} - (\mathbf{n}_{\text{OOP}}^T \cdot \boldsymbol{\sigma} \cdot \mathbf{n}_{\text{OOP}}) \mathbf{n}_{\text{OOP}}, \quad (11.1)$$

$$\mathbf{d}_{\text{OOP}} = \frac{\mathbf{D}_{\text{OOP}}}{|\mathbf{D}_{\text{OOP}}|}, \quad (11.2)$$

The subsequent calculation of ΔCFS on the OOP follows the same procedure as for the specified oriented planes with fixed focal mechanisms, described in Equations (3)–(4). It is important to note that the optimization targets CFS rather than ΔCFS , so the unit slip vector $\mathbf{d}_{\text{OOP}}$ is dominated by the full stress tensor $\boldsymbol{\sigma}$. In the dynamic case, the stress at each field point varies with time, so the OOP is the most failure-prone fault plane at each time point, and consequently both the OOP normal vector and slip vector are time-dependent. If the earthquake-induced stress perturbation ($\Delta\sigma_e$) is negligible relative to the tectonic stress field (i.e., $\boldsymbol{\sigma} \approx \boldsymbol{\sigma}_t$), or when the $\boldsymbol{\sigma}_t$ inferred from geological surveys represents postseismic conditions (i.e., $\boldsymbol{\sigma} = \boldsymbol{\sigma}_t$), then the

OOP is governed entirely by σ_t . Because only the OOP normal and the slip direction on the OOP are required, it suffices to know the principal stress orientations of σ_t ; their magnitudes are unnecessary. By contrast, when computing the optimal slip direction on a fault plane with prescribed strike and dip, one simply replaces the normal vector in Equation (11.1) with that of the plane. In this case, the full components of σ_t , that is, the complete stress tensor, must be provided.

3 Verification

We verified the key steps in ΔCFS calculation to ensure the reliability of the DynCFS program. As a test case, we specified a source in the AK135-FC model (Kennett *et al.* 1995; Montagner & Kennett 1996), with a fault size of 1 km $\times$ 1 km, a uniform slip of 1 m, and a centroid depth at 10 km, corresponding to an M_w 4.9 event. The source mechanism was set to match that of the 2008 M_w 7.9 Wenchuan earthquake (Y. Zhang *et al.* 2009; Y. Zhang 2010): (strike, dip, rake) = (223°, 47°, 131°). A receiver was then placed at an azimuth of 20°, an epicentral distance of 10 km, and a depth of 1 km. In our calculations, we sampled the Green's function library every 0.25 s, limited the upper frequency to the Nyquist limit, and used a normalized sine squared source time function (period = duration = 1.25 s) as a proxy for the δ function. The waveforms of the stress change components at the receiver, computed with QSEIS and QSSP and low-pass filtered at 0.4 Hz, are shown in Fig. 3, and the corresponding static stress changes in the ENZ coordinate system are (in Pa):

$$\Delta\sigma_e = \begin{bmatrix} 2603 & -2029 & -545 \\ -2029 & -2557 & -1671 \\ -545 & -1671 & 185 \end{bmatrix}. \quad (12)$$

Subsequently, unless explicitly stated otherwise, all coordinate systems employ the ENZ convention with tension taken as positive. The dynamic stress components computed with QSEIS and QSSP are nearly identical, and their final values both converge to those calculated using EDGRN–EDCMP. This confirms that our procedure, building dynamic and static Green's function libraries, reading stress components, and applying rotational superposition, is correct.

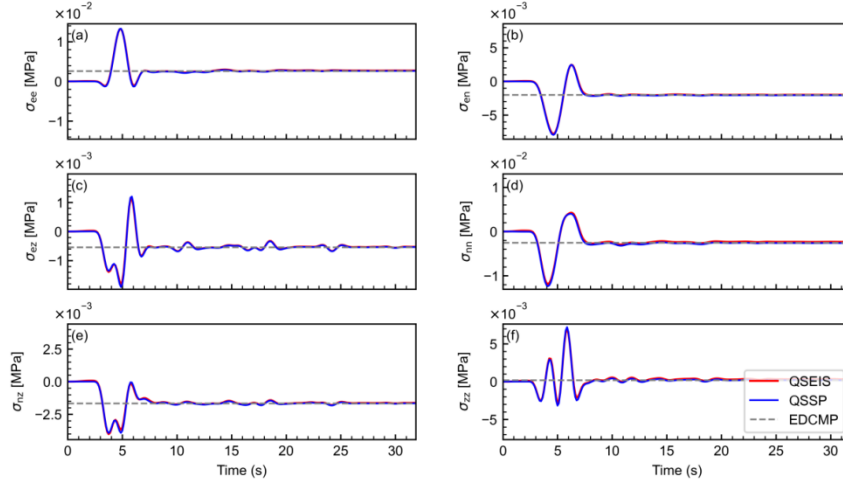


Figure 3. Comparison of stress components calculated by QSEIS, QSSP, and EDGRN-EDCMP in the AK135-FC Earth model. The source has a mechanism of (strike, dip, rake) = (223°, 47°, 131°), a moment magnitude of M_w 4.9, and a centroid depth of 10 km. The receiver is 10 km away, at an azimuth of 20° and a depth of 1 km.

We next validated the calculation of static ΔCFS for a specified receiver fault mechanism, and compared the obtained results with Coulomb3. With the slip model of the Wenchuan earthquake (Y. Zhang *et al.* 2009; Y. Zhang 2010), we computed the static ΔCFS on receivers at 15 km depth with the same mechanism as the main event. In a homogeneous half-space, we assigned elastic properties equal to those of the AK135-FC model (Kennett *et al.* 1995; Montagner & Kennett 1996) at the source depth: Poisson's ratio 0.18 and Young's modulus 7.3×10^{10} Pa. We further assumed pore pressure change proportional to the normal stress change and used an apparent friction coefficient $\mu' = 0.4$. Figures 4a-b compare results of our program with those from Coulomb3, showing almost identical spatial patterns. The layered half-space solution is shown in Fig. 4c. To illustrate the differences between uniform half-space and layered media, we then computed the pointwise relative difference between the results shown in Figs. 4a and 4c:

$$\Delta_R = \frac{|\Delta CFS_{uni} - \Delta CFS_{lay}|}{\max(|\Delta CFS_{uni}|, |\Delta CFS_{lay}|)}, \quad (13)$$

where Δ_R denotes the relative difference. As shown in Fig. 4a-c, the absolute differences between the uniform and layered media are small, whereas the Δ_R can be significant in some regions, particularly at far-field distances where both models suggest very weak stresses.

Because dynamic Δ CFS is computed at each time step using the same procedure as the static case, the combined results of Figs. 3 and 4 support the reliability of our dynamic Δ CFS estimates on receiver planes with specified mechanism.

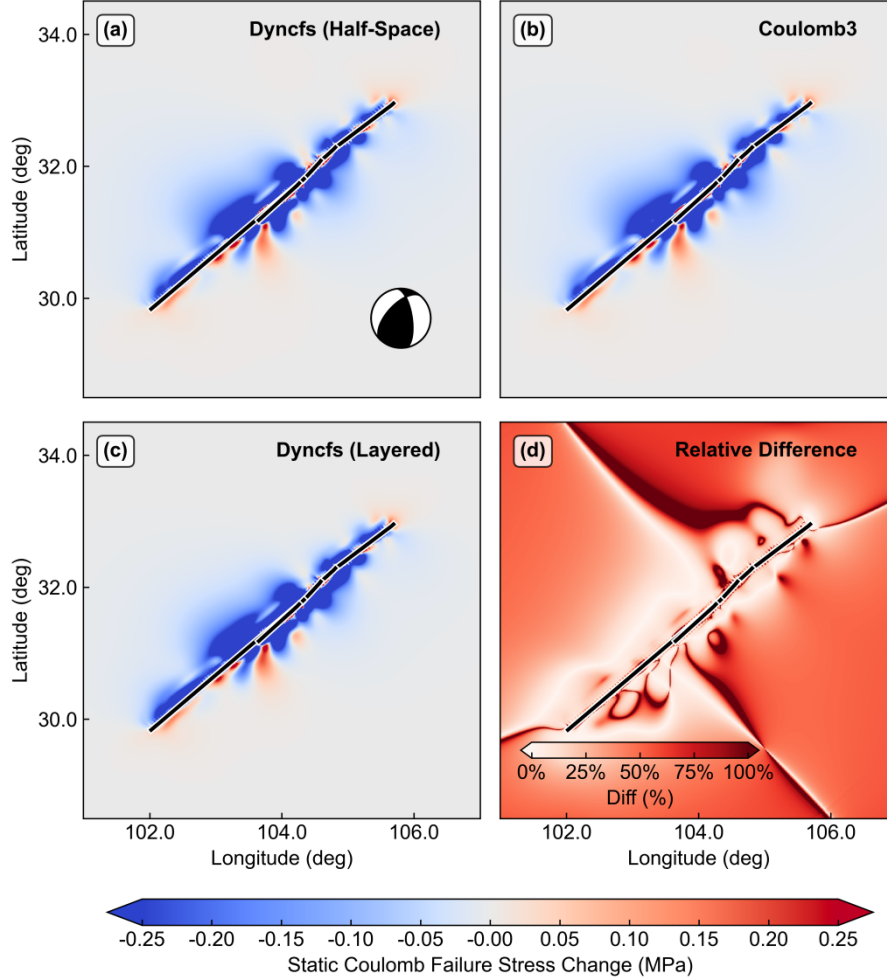


Figure 4. Static Coulomb failure stress change (Δ CFS) at 15 km depth cause by the 2008 Wenchuan earthquake. The earthquake mechanism was used as those of receiver faults. The thick black solid line marks the intersection between the main fault plane and the horizontal plane at 15 km depth. **(a)** Results from the DynCFS in homogeneous half-space. **(b)** Results from Coulomb3 in the same homogeneous half-space. **(c)** Results from DynCFS in layered elastic half-space. **(d)** Relative difference defined by Equation (13) between **(a)** and **(c)**.

We then computed the static stress change using $\mu = 0.8$ and $B = 0.5$ for the same source and receiver as in Fig. 3 and superimposed a regional deviatoric tectonic stress on the shallow region of Longmen Shan Fault. This stress field was derived from borehole hydraulic

fracturing stress data (Q. An et al. 2004; Y. Shi & J. Cao 2010). The minimum horizontal principal stress σ_1 (i.e., the maximum horizontal compressive stress) is oriented N60°W with a magnitude of -7.5 MPa; the maximum horizontal principal stress σ_2 (i.e., the minimum horizontal compressive stress) is orthogonal to it with a magnitude of -3 MPa; the vertical deviatoric stress σ_3 is approximately zero. Since the lithostatic pressure does not generate shear stress and produce identical normal stress and pore pressure on any fault plane, they have no effect on OOP selection and only influence the absolute magnitude of CFS. Therefore, the tectonic stress employed here represents deviatoric stress, and the subsequent CFS values also exclude contributions from lithostatic pressures. We performed a grid search over strike, dip, and rake at 1° increments and recorded the CFS for each combination. Fig. 5a displays the maximum CFS among all possible slip directions (rake angles) for each fault plane defined by specific strike and dip combinations. The fault planes corresponding to the two highest CFS values represent the OOPs. Both OOPs exhibit identical dip angles of 26° and rake angles of 90°, which is consistent with horizontal compression in the imposed tectonic stress field. The two OOPs have strikes of 30° and 210°, respectively, both perpendicular to the maximum horizontal compressive principal stress axis. The source used here, equivalent to an M_w 4.9 earthquake, produces a stress change $\Delta\sigma_e$ (Equation 12) negligible compared with the background deviatoric tectonic stress $\Delta\sigma_t^{\text{dev}}$. Consequently, the OOP orientation is governed almost entirely by the tectonic stress field. The two OOPs identified by the search have unit normal and slip vectors $\mathbf{n}_1 = [-0.22, 0.38, -0.90]^T$, $\mathbf{d}_1 = [0.45, -0.78, -0.44]^T$, and $\mathbf{n}_2 = [0.22, -0.38, -0.90]^T$, $\mathbf{d}_2 = [-0.45, 0.78, -0.44]^T$, respectively, which are in agreement with the results computed analytically using Equations (9)–(11). It is worth noting that, when determining the OOP, the quantity to be maximized should be CFS rather than ΔCFS . In Fig. 5b, we plot the static ΔCFS on the OOP (grey horizontal dashed line), which shows that the static ΔCFS is small compared with the total CFS, because the total CFS is the summation of the coseismic stress perturbation and the pre-existing tectonic stress. At the same time, we also plot the dynamic ΔCFS on the OOP at each time step in Fig. 5b (black curve), whose peak-to-peak amplitude is 6895 Pa, much larger in magnitude than the static value of -574 Pa.

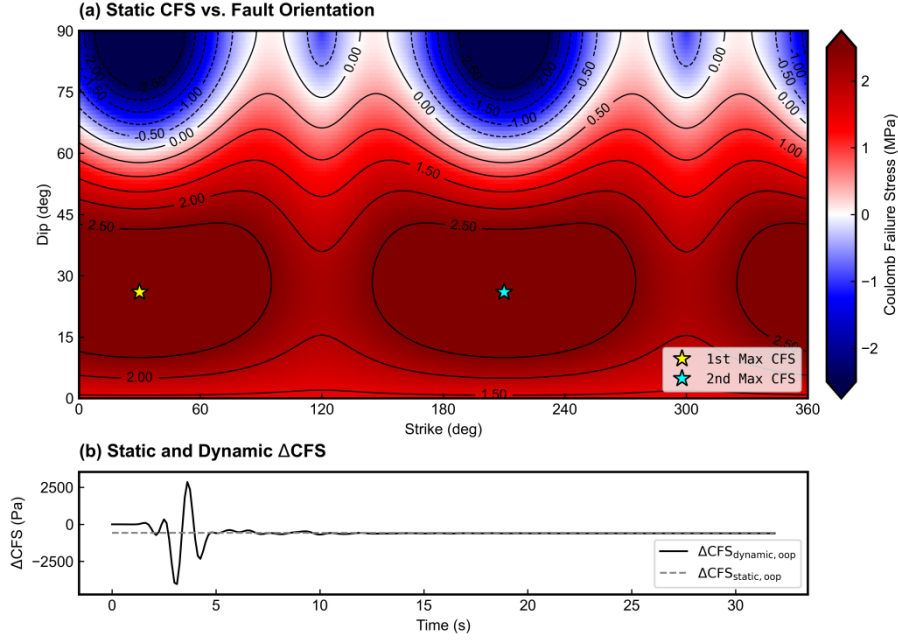


Figure 5. (a) Static CFS distribution relative to fault orientation. Both strike and dip are sampled at 1° intervals. The two stars mark the two OOPs, corresponding to CFS maxima of 3.20 MPa, with (strike, dip, rake) = $(30^\circ, 26^\circ, 90^\circ)$ and $(210^\circ, 26^\circ, 90^\circ)$, respectively. (b) Static Δ CFS on OOP (gray dashed line) and Dynamic Δ CFS evolution on time-dependent OOPs (black curve).

4 Application: 2008 M_W 7.9 Wenchuan Earthquake

The 2008 M_W 7.9 Wenchuan earthquake ruptured the Longmenshan fault system at the eastern margin of the Tibetan Plateau, where NW-dipping, imbricate reverse–oblique faults accommodate convergence against the rigid Sichuan Basin (Burchfiel *et al.* 2008; Y. Zhang *et al.* 2009). Geological, geodetic and seismological studies indicate multi-segment ruptures with mixed reverse and right-lateral slip on both the Yingxiu–Beichuan Fault (YBF) and the Pengguan Fault (PGF) (X. Xu *et al.* 2009; Y. Li *et al.* 2010; P. Zhang *et al.* 2010; Hartzell *et al.* 2013). The regional topography and traces of the Longmenshan system used in this study are shown in Fig. 6a. Guided by these constraints, Y. Zhang (2010) parameterized a multi-plane, mildly listric geometry including the YBF (planes S1–S4) and the PGF (plane S5), and then performed a joint finite-fault inversion that fits teleseismic, high-rate GNSS, and InSAR data. The source time function and slip distribution are shown in Fig. 6b.

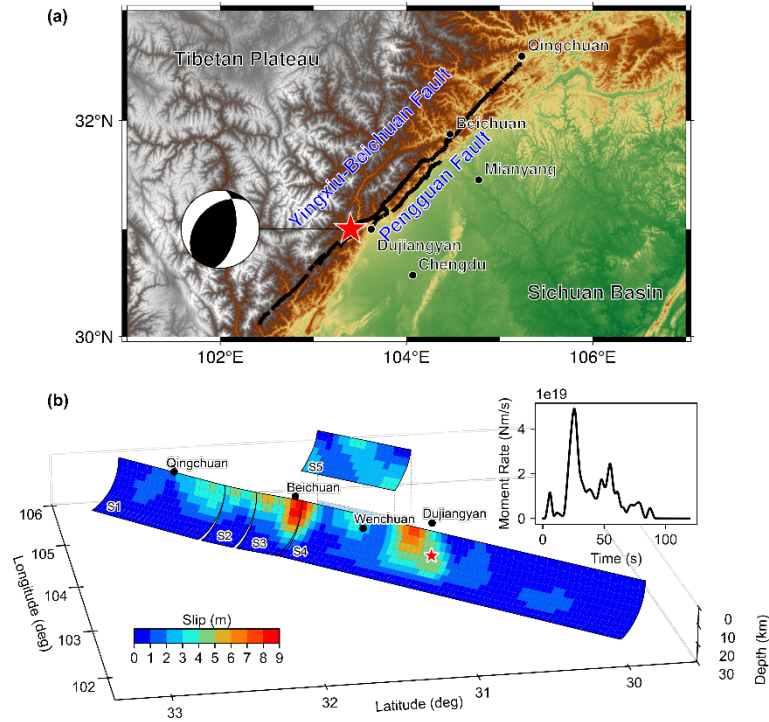


Figure 6. (a) Topography of the Longmenshan fault zone. The Yingxiu–Beichuan Fault (YBF) and the Pengguan Fault (PGF) are labeled, and the epicenter is marked with a red star. **(b)** Source time function and slip distribution in the 2008 M_w 7.9 Wenchuan Earthquake. Fault segments S1–S4 correspond to the YBF, and S5 corresponds to the PGF (Y. Zhang, 2010).

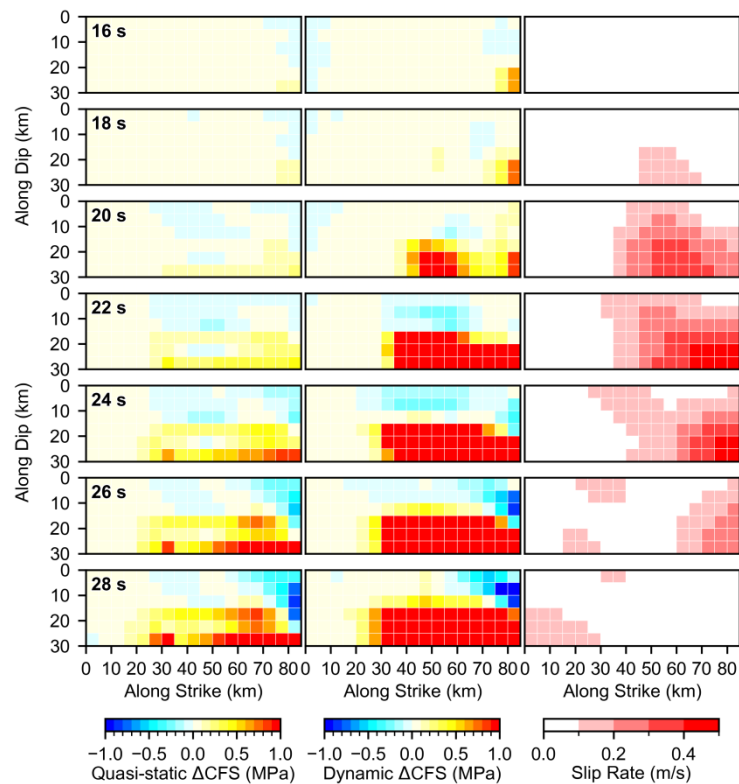


Figure 7. Columns from left to right show the temporal evolution of (left) quasi-static ΔCFS on segment S5 caused by the cumulative slip on segments S1–S4, (middle) fully dynamic ΔCFS on S5 including wave-propagation effects, and (right) slip rate (Y. Zhang, 2010) on S5.

Using the AK135-FC Earth model, we built a Green’s function library tailored to the Wenchuan earthquake. We first constructed the static stress Green’s functions. The library spans epicentral distances out to 1000 km at a 2 km interval; source depths from 1 to 31 km at a 2 km interval; and receiver depths from 0 to 32 km at a 2 km interval. Using this library and Equation (2), and assuming an effective friction coefficient $\mu' = 0.4$, we computed at each time the quasi-static ΔCFS on S5 caused by the cumulative slip on S1–S4; these results are shown in the left column of Fig. 7. This can be regarded as the time evolution of ΔCFS for a quasi-static rupture (i.e., zero rupture velocity and infinite seismic wave velocities). We then computed a dynamic stress Green’s function library with the same spatial sampling. Because the event is large and the finite-fault inversion used 5 km subfaults with an upper frequency of 0.2 Hz, we calculated dynamic stress Green’s functions using a 0.5 s time step and 1024 samples. And a normalized sine squared function (period = duration = 2.5 s) was set as a proxy for the δ function. Based on this library and Equation (2), we computed the time dependent dynamic ΔCFS caused by slips on S1–S4 to the receiver fault S5, again assuming $\mu' = 0.4$. The resulting ΔCFS evolution is shown in the middle column of Fig. 7, while the right column plots the slip rate history on S5. The dynamic ΔCFS substantially exceeds the static one at nearly all time points. Approximately 18 s after rupture onset, S5 initiated rapid failure below ~ 15 km depth and largely ceased by ~ 28 s, accumulating a total slip of ~ 3.19 m. The nucleation and subsequent growth patches of fast slip closely coincide with regions where dynamic ΔCFS reaches ~ 1 MPa, indicating that the YBF may have dynamically triggered the PGF.

5 Application: 2014 M_w 6.1 Ludian Earthquake

The 3 August 2014 M_w 6.1 Ludian earthquake struck the southeastern margin of the Tibetan Plateau, near the boundary with the stable South China Block, along the conjugate Zhaotong–Ludian and Baogunao–Xiaohe faults (Y. Zhang *et al.* 2015; Y. Luo *et al.* 2018; Y. Wan *et al.* 2024). This event is widely interpreted as rupture that occurred sequentially on two conjugate

fault planes (G. Zhang *et al.* 2014; Y. Wan *et al.* 2024). Guided by this view, Y. Zhang *et al.* (2015) performed a finite-fault inversion using both local and teleseismic waveform data on two conjugated planes; the resulting slip distribution is projected onto the surface in Fig. 8, together with the regional topography, focal-mechanism solution, and stress field.

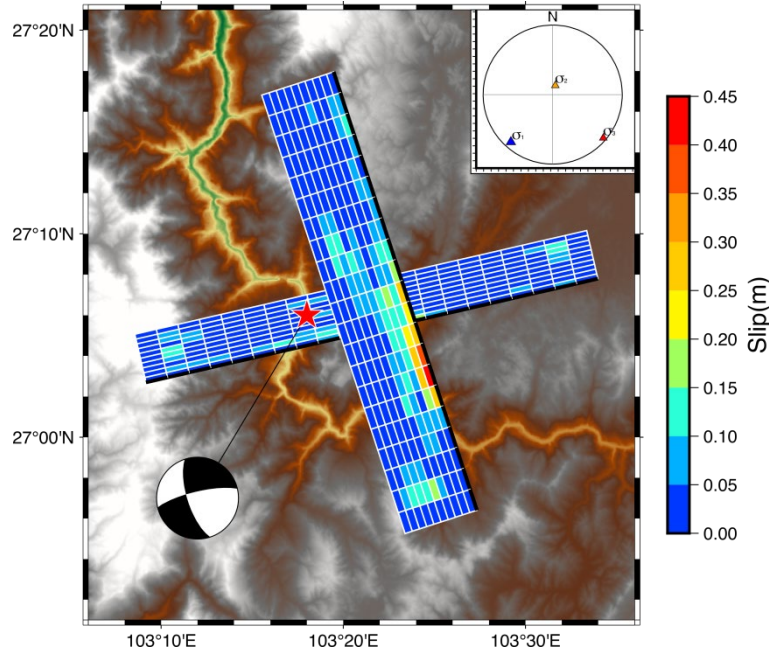


Figure 8. Regional topography and surface projection of slip distribution for the 2014 Ludian earthquake (Y. Zhang *et al.* 2015). The epicenter is shown by a red star and the focal-mechanism (black beachball) is from G. Zhang *et al.* (2014). The upper-right inset shows the lower-hemisphere projection of the regional stress field (Y. Wan *et al.* 2024).

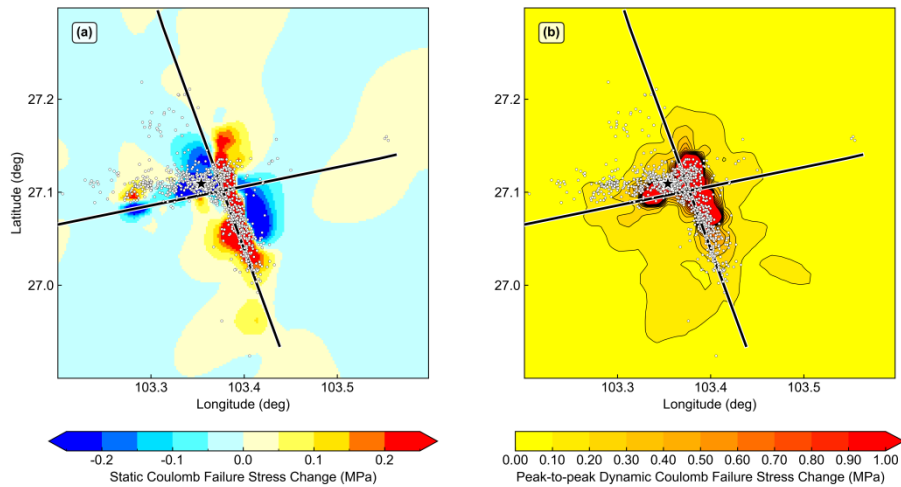


Figure 9. Coulomb failure stress changes on optimally oriented planes at 5 km depth and aftershocks (white circles) for the 2014 M_w 6.1 Ludian earthquake. Bold black lines mark the

fault positions at 5 km depth. Black star denotes the epicenter. (a) Static ΔCFS . (b) Peak-to-peak dynamic ΔCFS resolved on the OOPs during rupture.

Because the Ludian earthquake is smaller, we adopted finer space–time sampling for the Green’s function library. We set both the static and dynamic Green’s functions to a maximum epicentral distance of 150 km sampled every 0.25 km; source depths from 0 to 20 km with a 0.5 km interval; and receiver depths from 0.25 to 20.25 km with a 0.5 km spacing. Accordingly, for the dynamic Green’s functions we used a 0.125 s time step with 1024 samples. We again used a normalized sine squared function (period = duration = 0.725 s) as a proxy for the δ function. Finite-fault inversion indicates that slip was concentrated above ~ 5 km on the NNW–SSE–striking fault plane, with aftershocks largely confined to depths shallower than ~ 10 km. Guided by these constraints, we computed the static ΔCFS on the OOP at 5 km depth (Fig. 9a) assuming $\mu = 0.6$ and $B = 0.75$, using the regional tectonic stress field (Y. Wan *et al.* 2024). Meanwhile, we computed, at the same depth and over the same spatial domain, the dynamic stress field generated by the earthquake at each time step. Using the same regional stress field, we then determined, at each time step and for each observation point, the OOPs and the corresponding dynamic ΔCFS resolved on those OOPs. Subsequently, we evaluated the peak-to-peak dynamic ΔCFS over the entire sequence of time steps (defined as the maximum over all times minus the minimum over all times), as shown in Fig. 9b. We found that near the ENE–WSW–striking fault, the aftershocks fall within the static ΔCFS shadow (negative values) area, but correlate strongly with the peak-to-peak dynamic ΔCFS . This pattern suggests that dynamic ΔCFS can trigger a subset of aftershocks or, equivalently, that larger peak dynamic stresses produce greater clock advance (J. Gomberg *et al.* 1998). A similar behavior, in which peak ΔCFS aligns better with aftershock distribution than the static term, was documented for the 1992 M_w 7.3 Landers earthquake (Deborah Kilb *et al.* 2000). One interpretation is that dynamic stress perturbation effectively modifies the failure criterion and/or the characteristics and rates of rupture-promoting processes (Deborah Kilb *et al.* 2000; Debi Kilb *et al.* 2002).

6 Discussion

Unlike the static ΔCFS calculation, which accounts for the finite fault size or area, the dynamic

402 Δ CFS calculation treats each source as a point. To ensure computational accuracy, the source
 403 dimension must be reasonably small, particularly at near-field distances. In our applications to
 404 the 2008 Wenchuan and 2014 Ludian earthquakes, the subfault dimensions were neglected in
 405 the dynamic Δ CFS computations. To evaluate the discretization error introduced by this point
 406 approximation, we first consider a 5 km $\times$ 5 km subfault centered at 10 km depth with focal
 407 mechanism (strike, dip, rake) = (0°, 90°, 0°). We then uniformly partition it into 4, 9, 16, and
 408 25 smaller subfaults. For each discretization level, we compute, in a homogeneous elastic half-
 409 space, the six independent stress components at receivers located on azimuth 45°, at 1 km depth,
 410 for epicentral distances from 0 to 100 km. In this test we do not employ a precomputed Green's
 411 function library; instead, exact source–receiver stress waveforms are calculated directly, with
 412 an upper frequency of 0.2 Hz. Taking the arithmetic mean of the 25-subfault solution as the
 413 reference, the misfit is defined as the normalized RMS difference

$$\text{misfit} = \frac{\sum_i [\sigma(t_i) - \sigma_{\text{ref}}(t_i)]^2}{\sum_i [\sigma_{\text{ref}}(t_i)]^2}, \quad (14)$$

415 where $\sigma(t_i)$ and $\sigma_{\text{ref}}(t_i)$ denote the stress waveform and the reference waveform at time index i ,
 416 respectively. At an epicentral distance of 15 km, the single-point approximation yields a misfit
 417 of ~ 0.05 and a cross-correlation coefficient of ~ 0.98 (Fig. 10), indicating that discretization
 418 errors are negligible under these conditions. In Fig. 10c, we plot the σ_{ee} waveforms at an
 419 epicentral distance of 15 km for numbers of discretization points ranging from 1 to 25 as
 420 examples. The colors correspond to those in Figs 10a and 10b. It can be seen that the waveforms
 421 for different discretization points are already very similar. The corresponding waveforms for a
 422 larger epicentral distance at 50 km are shown in Fig. 10d, where the curves of different
 423 discretization points are almost identical. Generally, for epicentral distances exceeding about
 424 three times the subfault dimension, discretization errors are very weak, which can be neglected.

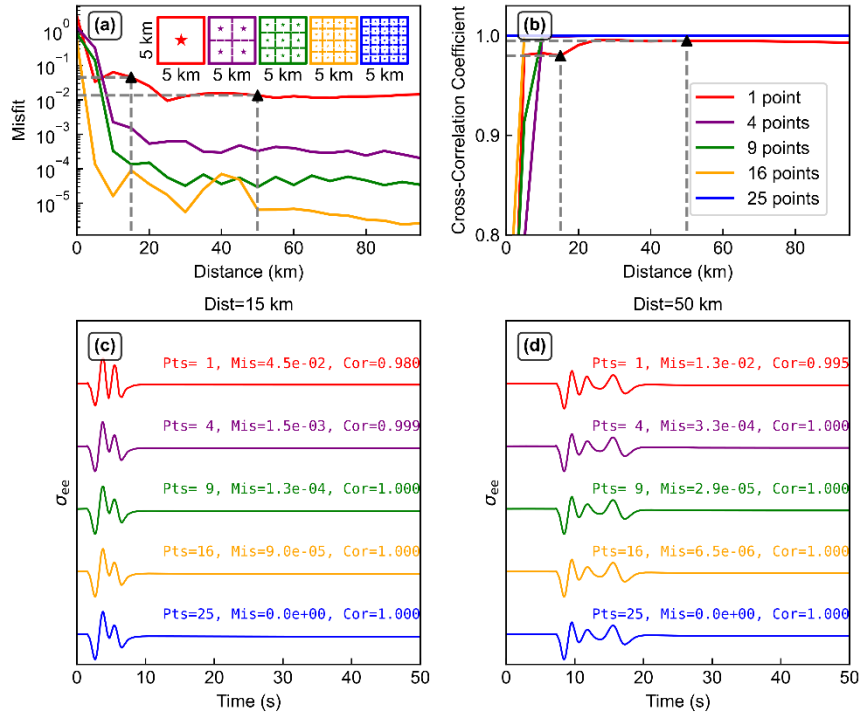


Figure 10. Discretization errors associated with subfault sampling. A 5 km $\times$ 5 km subfault is represented either by a single point source or by 4, 9, 16, and 25 uniformly distributed points (schematic in panel a). All points share the same focal mechanism, (strike, dip, rake) = (0°, 90°, 0°). The subfault center lies at 10 km depth; the receiver is located northeast of the subfault center (azimuth 45°) at a depth of 1 km. Waveform differences are evaluated relative to the highest sampling (25 points) as epicentral distance increases. (a) Misfit versus epicentral distance. (b) Cross-correlation coefficient versus epicentral distance. (c) σ_{ee} stress component for different numbers of discretization points at an epicentral distance of 15 km. (d) σ_{ee} stress component for different numbers of discretization points at an epicentral distance of 50 km.

7 Conclusion

We developed DynCFS, an open-source Python-based software tool to compute both static and dynamic Coulomb failure stress changes (Δ CFS) in a layered elastic half-space. The code efficiently evaluates the dynamic Δ CFS generated by a finite-fault source on (i) receiver faults with a prescribed mechanism, (ii) receiver faults with fixed geometry but rake optimized with respect to a tectonic stress field, and (iii) receiver faults for which both geometry and rake are optimized under a tectonic stress field.

DynCFS offers several advantages over existing static Coulomb stress tools and fully 3D numerical simulations. First, it provides a unified framework to calculate both static and dynamic Δ CFS with the same layered elastic structure, allowing direct comparison between static and dynamic triggering effects. Second, it employs a suite of techniques developed by R. Wang et al. (1999, 2007, 2017) to suppress numerical errors, which greatly reduces numerical inaccuracies compared with the conventional propagator-matrix and normal-mode summation approaches. Third, by supporting both user-specified receiver planes and optimally oriented planes, the code can be used to evaluate stress changes on mapped faults as well as on potential failure planes consistent with the regional stress field. Fourth, because it relies on 1D layered Green's functions rather than full 3D meshes, DynCFS is computationally inexpensive and easy to apply to many source–receiver configurations, making it suitable for systematic parameter studies and routine applications. Moreover, in the near field, computing stress waveforms using normal-mode summation would require very high spherical-harmonic degrees; by instead adopting a reasonable horizontally layered-medium approximation and employing cylindrical-harmonic summation, we greatly accelerate the computation of the Green's function library. In typical applications, full simulations for a large-size finite-fault model and a dense grid of receiver points can be completed within a few hours on a mid-range 2021-era desktop processor (e.g., an AMD Ryzen 7 5800X). Finally, the implementation in Python and the open-source distribution lower the barrier to use.

We demonstrated the performance and usefulness of DynCFS with applications to the 2008 M_W 7.9 Wenchuan and 2014 M_W 6.1 Ludian earthquakes. For the Wenchuan earthquake, the calculations illuminate the spatial and temporal processes of Δ CFS on secondary fault segments and support a dynamic triggering interpretation of the rupture of PGF (segment S5). For the Ludian earthquake, the results show that regions predicted to be static stress shadows can still experience positive dynamic Δ CFS that correlate with the aftershock distribution, underscoring the importance of dynamic effects. These examples illustrate that DynCFS is a practical tool for studying rupture processes and stress transfer, and for quantifying dynamic triggering in layered media structures.

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Data Availability Statement

The DynCFS program and two companion applications are open-sourced on GitHub: <https://github.com/Zhou-Jiangcheng/dyncfs>. A brief installation and usage guide is provided in the repository's README.md, while detailed parameter descriptions are documented in the ini files for the two applications under the example's directory.

Complicity of Interests

The authors declare that they have no conflict of interest in relation to this study.

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Appendix 1. Procedure for obtaining each stress component

When solving the wave equation, we follow the orthogonal normalization propagator matrix method used in QSEIS (R. Wang, 1999) and can optionally employ the differential transform to suppress numerical errors (R. Wang & H. Wang, 2017). Other formula derivations are modified from the notes of Dr. Wang Rongjiang (Unpublished). Considering the spatial distribution characteristics of a horizontally layered medium, the cylindrical coordinate system is an appropriate choice. We take the upward direction as the positive z -axis, the outward direction as the positive r -axis, and the counterclockwise direction as the positive θ -axis to establish a right-handed coordinate system. When separating spatial parameters by variables, the Hankel transform is adopted. The scalar cylindrical harmonic function in the transform is defined as

$$Y_k^m(r, \theta) = J_m(kr) e^{im\theta}, \quad (A1)$$

where k is the wavenumber, r is the epicentral distance, $J_m(kr)$ is the Bessel function of order m , and $Y_k^m(r, \theta)$ satisfies the following equations:

$$\begin{cases} \frac{\partial^2 Y_k^m(r, \theta)}{\partial r^2} + \frac{1}{r} \frac{\partial Y_k^m(r, \theta)}{\partial r} - \frac{m^2}{r^2} Y_k^m(r, \theta) = -k^2 Y_k^m(r, \theta) \\ \frac{\partial^2 Y_k^m(r, \theta)}{\partial \theta^2} = -m^2 Y_k^m(r, \theta) \end{cases}. \quad (A2)$$

Its partial derivatives with respect to r and θ are

$$\begin{cases} \frac{\partial Y_k^m(r, \theta)}{\partial r} = k J_m'(kr) e^{im\theta} \\ \frac{\partial Y_k^m(r, \theta)}{\partial \theta} = im J_m(kr) e^{im\theta} \end{cases}. \quad (A3)$$

Based on this scalar cylindrical harmonic function, the three basic functions of the vector Hankel transform can be given as

$$\begin{cases} \mathbf{Z}_k^m(r, \theta) = \mathbf{e}_z Y_k^m(r, \theta) \\ \mathbf{R}_k^m(r, \theta) = \left(\mathbf{e}_r \frac{\partial}{k \cdot \partial r} + \mathbf{e}_\theta \frac{\partial}{kr \cdot \partial \theta} \right) Y_k^m(r, \theta) \\ \mathbf{T}_k^m(r, \theta) = \left(\mathbf{e}_r \frac{\partial}{kr \cdot \partial \theta} - \mathbf{e}_\theta \frac{\partial}{k \cdot \partial r} \right) Y_k^m(r, \theta) \end{cases}. \quad (A4)$$

If there exists a vector

$$\mathbf{f} = U_m \mathbf{Z}_k^m(r, \theta) + V_m \mathbf{R}_k^m(r, \theta) + W_m \mathbf{T}_k^m(r, \theta) = f_z \mathbf{e}_z + f_r \mathbf{e}_r + f_\theta \mathbf{e}_\theta. \quad (A5)$$

then its cylindrical coordinate components can be obtained from (A2)–(A5) as

$$\begin{cases} f_z = U_m J_m(kr) e^{im\theta} \\ f_r = V_m J_m'(kr) e^{im\theta} + W_m \frac{im}{kr} J_m(kr) e^{im\theta} \\ f_\theta = V_m \frac{im}{kr} J_m(kr) e^{im\theta} - W_m J_m'(kr) e^{im\theta} \end{cases}. \quad (A6)$$

672 From the recurrence relations of the Bessel function

$$673 \quad J'_m(x) = \frac{1}{2} [J_{m-1}(x) - J_{m+1}(x)], \quad (\text{A7.1})$$

$$674 \quad J_m(x) = \frac{x}{2m} [J_{m-1}(x) + J_{m+1}(x)], \quad (\text{A7.2})$$

675 (A6) can be rewritten into a form more convenient for programming as

$$676 \quad \begin{cases} f_z = U_m J_m(kr) e^{im\theta} \\ f_r = \frac{1}{2} (V_m + iW_m) J_{m-1}(kr) e^{i(m-1)\theta} - \frac{1}{2} (V_m - iW_m) J_{m+1}(kr) e^{i(m+1)\theta} \\ f_\theta = \frac{i}{2} (V_m + iW_m) J_{m-1}(kr) e^{i(m-1)\theta} + \frac{i}{2} (V_m - iW_m) J_{m+1}(kr) e^{i(m+1)\theta} \end{cases} \quad (\text{A8})$$

677 When solving the wave equation in a horizontally layered medium using the propagator matrix
678 method, the boundary conditions are continuity of displacement and continuity of stress across
679 layers. Therefore, it is convenient to expand the displacement vector $\mathbf{u}(z, r, \theta)$ and the stress
680 tensor $\mathbf{\Gamma}(z, r, \theta)$ by the following inverse Hankel transforms:

$$681 \quad \begin{cases} \mathbf{u}(z, r, \theta) = \sum_m \int_0^\infty (U_m \mathbf{Z}_k^m(r, \theta) + V_m \mathbf{R}_k^m(r, \theta) + W_m \mathbf{T}_k^m(r, \theta)) k dk \\ \mathbf{e}_z \cdot \mathbf{\Gamma}(z, r, \theta) = \sum_m \int_0^\infty (E_m \mathbf{Z}_k^m(r, \theta) + F_m \mathbf{R}_k^m(r, \theta) + G_m \mathbf{T}_k^m(r, \theta)) k dk \end{cases} \quad (\text{A9})$$

682 so that the corresponding displacement–stress vector in the transform domain is

$$683 \quad \mathbf{y} = \begin{bmatrix} U_m \\ E_m \\ V_m \\ F_m \\ W_m \\ G_m \end{bmatrix}. \quad (\text{A10})$$

684 From (A8), the displacement–stress vector $\mathbf{y}$ can be transformed into cylindrical coordinates,
685 yielding the three displacement components and the projection of the stress tensor in the vertical
686 direction:

$$687 \quad \begin{bmatrix} u_z \\ \sigma_{zz} \\ u_r \\ \sigma_{zr} \\ u_\theta \\ \sigma_{z\theta} \end{bmatrix} = \begin{bmatrix} U_m J_m(kr) e^{im\theta} \\ E_m J_m(kr) e^{im\theta} \\ \frac{1}{2} (V_m + iW_m) J_{m-1}(kr) e^{i(m-1)\theta} - \frac{1}{2} (V_m - iW_m) J_{m+1}(kr) e^{i(m+1)\theta} \\ \frac{1}{2} (F_m + iG_m) J_{m-1}(kr) e^{i(m-1)\theta} - \frac{1}{2} (F_m - iG_m) J_{m+1}(kr) e^{i(m+1)\theta} \\ \frac{i}{2} (V_m + iW_m) J_{m-1}(kr) e^{i(m-1)\theta} + \frac{i}{2} (V_m - iW_m) J_{m+1}(kr) e^{i(m+1)\theta} \\ \frac{i}{2} (F_m + iG_m) J_{m-1}(kr) e^{i(m-1)\theta} + \frac{i}{2} (F_m - iG_m) J_{m+1}(kr) e^{i(m+1)\theta} \end{bmatrix}. \quad (\text{A11})$$

688 The order of the Bessel function depends only on the radiation pattern: explosive and CLVD
689 sources correspond to the 0th-order Bessel function, dip-slip and strike-slip sources correspond
690 to the 1st- and 2nd-order Bessel functions, respectively. Thus, we only need to compute the
691 following basic moment tensor components:

$$\mathbf{M}_{\text{exp}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (\text{A12.1})$$

$$\mathbf{M}_{\text{clvd}} = \begin{bmatrix} -0.5 & 0 & 0 \\ 0 & -0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (\text{A12.2})$$

$$\mathbf{M}_{\text{ds1}} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad (\text{A12.3})$$

$$\mathbf{M}_{\text{ss1}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad (\text{A12.4})$$

For a fixed azimuth θ_g (without loss of generality, let $\theta_g=0$) and a range of source depths $h_{s,g}$, receiver depths $h_{r,g}$, and epicentral distances r_g , the displacement–stress vector $\mathbf{y}_g$ serves as the Green’s function library. Specifically, for explosive and CLVD sources, the tangential mode components are zero, so only the radial and vertical components need to be stored, meaning a total of 10 Green’s function components. Thereafter, the physical quantities excited by any moment tensor source at any azimuth θ can be obtained by rotation and superposition from this Green’s function library. For a given combination of source depth h_s , receiver depth h_r , epicentral distance r , the corresponding quantities can be evaluated either by trilinear interpolation from neighboring Green’s functions or directly from the Green’s functions at the nearest grid point. In particular, for any P–SV mode physical quantity:

$$f_{\text{P-SV}}(r, \theta) = A_{\text{exp}} G_{\text{exp}}(r) + A_{\text{clvd}} G_{\text{clvd}}(r) + (A_{\text{ss1}} \sin 2\theta + A_{\text{ss2}} \cos 2\theta) G_{\text{ss}}(r) + (A_{\text{ds1}} \sin \theta + A_{\text{ds2}} \cos \theta) G_{\text{ds}}(r), \quad (\text{A13.1})$$

for any SH mode physical quantity:

$$f_{\text{SH}}(r, \theta) = (A_{\text{ss1}} \cos 2\theta - A_{\text{ss2}} \sin 2\theta) G_{\text{ss}}(r) + (A_{\text{ds1}} \sin \theta - A_{\text{ds2}} \cos \theta) G_{\text{ds}}(r), \quad (\text{A13.2})$$

where the coefficients are obtained from the given moment tensor $\mathbf{M}$ (in NED coordinates) as:

$$\begin{cases} A_{\text{exp}} = \frac{1}{3}(M_{11} + M_{22} + M_{33}) \\ A_{\text{clvd}} = -\frac{1}{3}(M_{11} + M_{22} - 2M_{33}) \\ A_{\text{ds1}} = M_{13}, \quad A_{\text{ds2}} = M_{23} \\ A_{\text{ss1}} = M_{12}, \quad A_{\text{ss2}} = \frac{1}{2}(M_{11} - M_{22}) \end{cases}. \quad (\text{A14})$$

The above describes the construction of the Green’s function library for displacement and stress projected in the z -direction, and the theoretical method for synthetic seismograms. In cylindrical coordinates, the strain components are:

$$\begin{aligned}
714 \quad & \left\{ \begin{array}{l} \varepsilon_{zz} = \frac{\partial u_z}{\partial z} \\ \varepsilon_{rr} = \frac{\partial u_r}{\partial r} \\ \varepsilon_{\theta\theta} = \frac{1}{r} \left(u_r + \frac{\partial u_\theta}{\partial \theta} \right) \\ \varepsilon_{zr} = \frac{1}{2} \left(\frac{\partial u_z}{\partial r} + \frac{\partial u_r}{\partial z} \right) \\ \varepsilon_{z\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z} \right) \\ \varepsilon_{r\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right) \end{array} \right. , \quad (A15)
\end{aligned}$$

715 For a homogeneous isotropic medium, the constitutive relation is:

$$716 \quad \sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij}. \quad (A16)$$

717 In the above, the projection of σ_{ij} in the z-direction has already been given in (A11), so we

718 only need to compute ε_{rr} , $\varepsilon_{\theta\theta}$ and $\varepsilon_{r\theta}$ from (A15). For $r \neq 0$, the results are:

$$\begin{aligned}
719 \quad & \varepsilon_{\theta\theta} = \frac{u_r}{r} + \frac{\partial u_\theta}{r \partial \theta} \\
& = -\frac{m-1}{2r} (V_m + iW_m) J_{m-1}(kr) e^{i(m-1)\theta}, \\
& \quad -\frac{m+1}{2r} (V_m - iW_m) J_{m+1}(kr) e^{i(m+1)\theta} \quad (A17.1)
\end{aligned}$$

$$720 \quad \varepsilon_{rr} = (\nabla \cdot \mathbf{u} - \varepsilon_{zz}) - \varepsilon_{\theta\theta} = -kV_m J_m(kr) e^{im\theta} - \varepsilon_{\theta\theta}, \quad (A17.2)$$

$$\begin{aligned}
& \varepsilon_{r\theta} = \frac{1}{2} \left(\frac{\partial u_\theta}{\partial r} + \frac{1}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r} \right) \\
& = \omega_z + \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r} \right) \\
721 \quad & = \frac{1}{2} (\nabla \times \mathbf{u}) \cdot \mathbf{e}_z + \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r} \right) \quad (A17.3) \\
& = \frac{1}{2} kW_m J_m(kr) e^{im\theta} + \frac{i(m-1)}{2r} (V_m + iW_m) J_{m-1}(kr) e^{i(m-1)\theta} \\
& \quad - \frac{i(m+1)}{2r} (V_m - iW_m) J_{m+1}(kr) e^{i(m+1)\theta}
\end{aligned}$$

722 When $r = 0$, if the source depth differs from the station depth, then all displacement and strain

723 components should converge. From the properties of $\lim_{r \rightarrow 0} J_m(kr)$ and the recurrence relation

724 (A7.2), we have, For $m = 0$:

$$725 \quad \varepsilon_{\theta\theta} = \frac{k}{4} (V_0 + iW_0) e^{-i\theta} - \frac{k}{4} (V_0 - iW_0) e^{i\theta}, \quad (A18.1)$$

$$726 \quad \varepsilon_{rr} = -kV_0 e^{i \cdot 0\theta} - \varepsilon_{\theta\theta}, \quad (A18.2)$$

$$727 \quad \varepsilon_{r\theta} = \frac{1}{2} kW_0 e^{i \cdot 0\theta} + \frac{ik}{4} (V_0 + iW_0) e^{-i\theta} + \frac{ik}{4} (V_0 - iW_0) e^{i\theta}. \quad (A18.3)$$

728 For $m = 1$,

$$729 \quad \varepsilon_{\theta\theta} = 0, \quad (A18.4)$$

$$730 \quad \varepsilon_{rr} = 0, \quad (A18.5)$$

$$731 \quad \varepsilon_{r\theta} = 0. \quad (A18.6)$$

732 For $m = 2$,

733
$$\varepsilon_{\theta\theta} = -\frac{k}{4}(V_2 + iW_2)e^{i\theta}, \quad (\text{A18.7})$$

734
$$\varepsilon_{rr} = -\varepsilon_{\theta\theta}, \quad (\text{A18.8})$$

735
$$\varepsilon_{r\theta} = \frac{ik}{4}(V_2 + iW_2)e^{i\theta}. \quad (\text{A18.9})$$

736